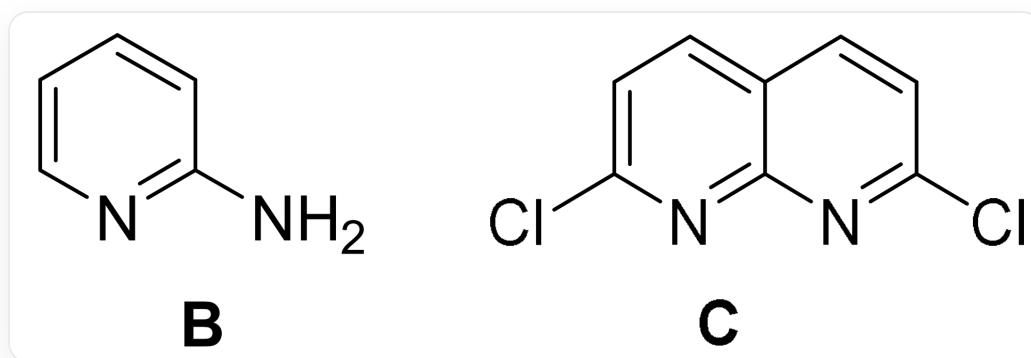


## Question

Due to the unique properties of metal-metal bonds, the synthesis of metal string complexes has consistently garnered widespread attention from researchers. A common method for synthesizing the ligand **A** for metal strings is as follows: Under an argon atmosphere, substrate **B** (1.98 g, 23 mmol) and **C** (2.16 g, 10 mmol),  $t$ -BuOK (2.81 g, 25 mmol),  $\text{Pd}_2(\text{dba})_3$  (183 mg, 0.20 mmol), dppp (165 mg, 0.40 mmol) were added to a dry flask. The mixture was stirred and refluxed in toluene (100 mL) for 48 hours, and the system was observed to turn dark brown. The solvent was removed under reduced pressure, and then the mixture was poured into water. The precipitate was filtered to obtain a solid, which was purified by silica gel column chromatography using ethyl acetate as the eluent to obtain a bright yellow product **A**. It is known that dppp is a bidentate phosphine ligand, and the structures of **B** and **C** are as follows:



The SMILES of **B** is NC1=CC=CC=N1, and the SMILES of **C** is ClC1=CC2=C(C=C1)C=CC(Cl)=N2

A mixture of **A** (251 mg), naphthalene (25 g), and DMF (5 mL) was placed in a conical flask and heated to 100 °C. Then  $t$ -BuOK (224 mg, 2.0 mmol) was slowly added, and the system immediately turned dark brown. It was heated at approximately 130~140 °C for 15 hours, and then refluxed at 200 °C for 15 minutes to obtain an orange transparent solution. Anhydrous  $\text{CoCl}_2$  (182 mg, 1.4 mmol) was added dropwise, and after refluxing for another 30 minutes,  $\text{NaSCN}$  (32 mg, 0.40 mmol) was slowly added, and the color changed from dark brown to dark green. The mixture was refluxed for another 3 hours, then concentrated to approximately 5 mL, and then cooled to 70 °C. It was treated with hexane, and the metal complex precipitated. The precipitate was collected by suction filtration and washed with hexane to remove naphthalene and DMF. The solid was extracted with  $\text{CH}_2\text{Cl}_2$  and filtered. The  $\text{CH}_2\text{Cl}_2$  solution was treated with excess  $\text{KPF}_6$  in methanol and stirred overnight, and then the solvent was removed under vacuum. The remaining solid was extracted with  $\text{CH}_2\text{Cl}_2$  and filtered. Pentane vapor was slowly diffused

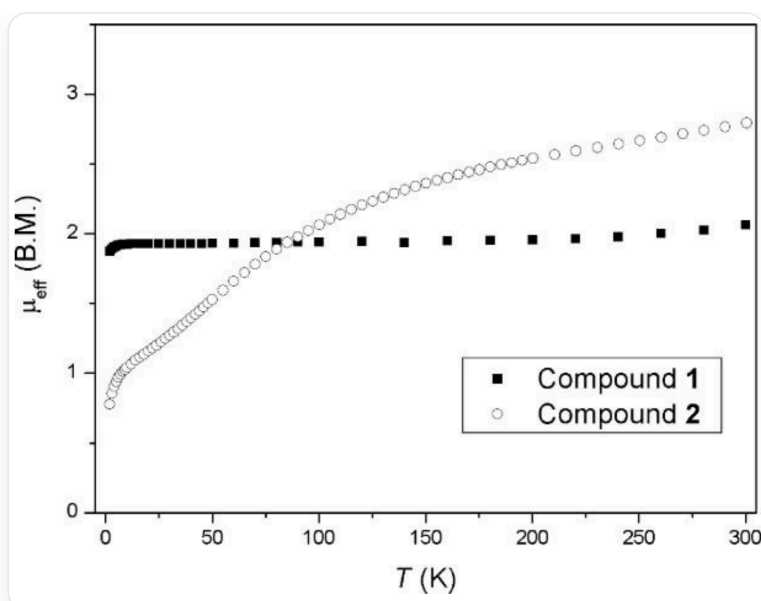
into the filtrate to obtain dark purple 1:1 salt **D**. The mass fractions of each element in **D** (all theoretical values) are as follows: Co 18.14%, C 46.22%, H 2.59%, N 18.69%. It is known that all coordinating atoms in the complex are the same.

**D** was treated with  $[\text{Cp}_2\text{Fe}](\text{PF}_6)$  in  $\text{CH}_2\text{Cl}_2/\text{MeOH}$  in a 1:1 reaction to obtain **E**.

At room temperature, the highest occupied single electron of **D** occupies  $\delta^*(6)$ ; the highest occupied two single electrons of **E** occupy  $\delta^*(6)$  and  $\sigma^*(4)$  with similar energy levels, respectively.

Which of the following statements are correct:

1. Given that the average bond order of the Co – Co bond in **D** is 0.5, then Co all satisfy the 18-electron rule.
2. There are 38 atoms involved in coordination in all anions and cations of **E**, and there are 2 types of coordinating atoms.
3. In the process of preparing **D**, a substance must have been oxidized in the reaction system.
4. Considering the distance between the farthest Co atoms, **D** > **E**
5. The magnetic moments of **D** and **E** vary with temperature as shown in the figure below:



This image is a two-dimensional scatter plot with lines against a white background. The vertical Y-axis is labeled " $\mu_{\text{eff}}$  (B.M.)", with a scale starting from 0 and incrementing upwards to 1, 2, and 3. The horizontal X-axis is labeled " $T$  (K)", with a scale starting from 0 and incrementing to the right with values 50, 100, 150, 200, 250, and 300. The plot shows two curves formed by discrete data points, as indicated in the legend: black squares represent compound **1**, and white circles represent compound **2**. The first curve, represented by solid black squares, has a  $\mu_{\text{eff}}$  value of approximately 1.9 in the very low temperature region near 0 K, which then remains essentially constant, forming a flat region up to about  $T = 75$  K. After 75 K, the  $\mu_{\text{eff}}$  value of this curve increases very slowly with increasing temperature, reaching about 2.1 at 300 K. The second curve, represented by hollow white circles, has a  $\mu_{\text{eff}}$  value below 1.0, around 0.8, near 0 K. As the temperature  $T$  increases, the  $\mu_{\text{eff}}$  value of this curve continues to rise significantly, with a higher rate of increase in the low-temperature region, which gradually slows down as the temperature increases. At approximately  $T = 80$  K, this hollow circle curve intersects with the solid square curve, where the  $\mu_{\text{eff}}$  value is slightly below 2.0. At  $T = 300$  K, the hollow circle curve reaches its highest point in the graph, with a  $\mu_{\text{eff}}$  value of approximately 2.8.

Then the black squares represent **E**, and the white circles represent **D**

**A.** All other options are incorrect

**B.** 1,3,4

**C.** 2,3,5

**D.** 3,4

**E.** 3,4,5

**F.** 1,3

**G.** 2,4

**H.** 4,5

**I.** 1,3,4,5

**J.** 2,5

**K.** 3,4,5

**L.** 2

**M.** 1,3,5

**N.** 2,3,4,5

**O.** 4

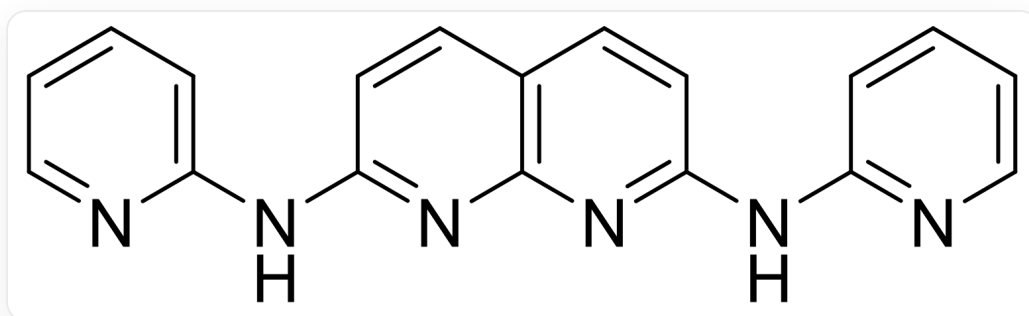
**P.** 2,3,4

## Answer

Correct Answer: **P**

## Detailed Explanation

**B** and **C** react, undergoing two couplings to obtain **A** :



SMILES为 C1(C=CC(NC2=CC=CC=N2)=N3)=C3N=C(NC4=NC=CC=C4)C=C1

### CHECKPOINT

1 PTS

**A** is C1(C=CC(NC2=CC=CC=N2)=N3)=C3N=C(NC4=NC=CC=C4)C=C

The anion of **A** corresponds to the ligand  $L^{2-}$ . From the mass fraction,  $n(C)/n(Co) = 12.50$ ,  $n(H)/n(Co) = 8.35$ ,  $n(N)/n(Co) = 4.33$ , and combined with the structure of the ligand  $L^{2-}$ , there should be 6 Co atoms on a metal string (**A**).

### CHECKPOINT

1 PTS

There should be 6 Co atoms on a metal string

From the mass fraction, we know  $n(\text{C}) = 75$ ,  $n(\text{H}) = 50$ ,  $n(\text{N}) = 26$

#### CHECKPOINT

1 PTS

$$n(\text{C}) = 75, n(\text{H}) = 50, n(\text{N}) = 26$$

From the number of N, it can be seen that **D** contains 4  $\text{L}^{2-}$  and 2  $\text{SCN}^-$ , containing a total of 74 C and 48 H, with an excess of 1 C atom and 2 H atoms.

The anion of the compound should be  $\text{PF}_6^-$ . Assuming there is only 1 anion, the remainder is:  
 $75M(\text{C})/46.22\% - 6M(\text{Co}) - 4M(\text{L}^{2-}) - 2M(\text{SCN}^-) - M(\text{PF}_6^-) = 84.76 \text{ g/mol}$

#### CHECKPOINT

2 PTS

Assuming there is only 1 anion, the remainder of **D** is 84.76 g/mol

This corresponds exactly to  $\text{CH}_2\text{Cl}_2$ . Combined with the excess of 1 C atom and 2 H atoms, the complex has one solvated molecule of  $\text{CH}_2\text{Cl}_2$ .

All coordinating atoms are the same, so it can be seen that  $\text{SCN}^-$  should coordinate with the N atom.

Therefore, the chemical formula of **D** is  $[\text{Co}_6(\mu_6 - \text{L})_4(\text{NCS})_2](\text{PF}_6) \cdot \text{CH}_2\text{Cl}_2$

#### CHECKPOINT

1 PTS

The chemical formula of **D** is  $[\text{Co}_6(\mu_6 - \text{L})_4(\text{NCS})_2](\text{PF}_6) \cdot \text{CH}_2\text{Cl}_2$

The metal core in **D** is  $\text{Co}_6^{11+}$ .  $\text{Co}^{2+}$  is reduced, so a substance must be oxidized in the reaction system. Statement 3 is correct. Counting the electrons provided by the 26 N atoms and the 5 Co – Co bonds with a bond order of 0.5, a total of  $6 \times 9 - 11 + 26 \times 2 + 5 \times 0.5 \times 2 = 100 < 6 \times 18$ . Statement 1 is incorrect.

## CHECKPOINT

1 PTS

In **D**, 6Co has a total of 100 electrons, and it is impossible to satisfy the 18-electron rule for all of them

## CHECKPOINT

0.5 PTS

A substance must be oxidized in the reaction system for the synthesis of **D**

**E** is **D** oxidized by a single electron, i.e.,  $[\text{Co}_6(\mu_6 - \text{L})_4(\text{NCS})_2](\text{PF}_6)_2 \cdot 2\text{CH}_2\text{Cl}_2$  (the coordinating solvent is not required and does not affect the judgment of the options). There are 26 N in the cation and 12 F in the anion participating in coordination. Statement 2 is correct. The average oxidation state of Co in **E** is higher, so the Co – Co bond order is larger to alleviate the electron deficiency, and Co – Co is shorter. Statement 4 is correct.

## CHECKPOINT

0.5 PTS

**E** is  $[\text{Co}_6(\mu_6 - \text{L})_4(\text{NCS})_2](\text{PF}_6)_2 \cdot 2\text{CH}_2\text{Cl}_2$

## CHECKPOINT

1 PTS

Compared to **D**, the average oxidation state of Co in **E** is higher, and Co – Co is shorter

The magnetic moment of **D** with 1 unpaired electron is basically unaffected by temperature changes. While the electrons of **E** can be excited as the temperature rises, transitioning between the singlet/triplet state. Therefore, the correct answer is: the black square corresponds to **D**, and the white circle corresponds to **E**. Statement 5 is incorrect.

**CHECKPOINT**

1 PTS

The black square corresponds to **D**, and the white circle corresponds to **E**

Statements 2, 3, and 4 are correct, choose P